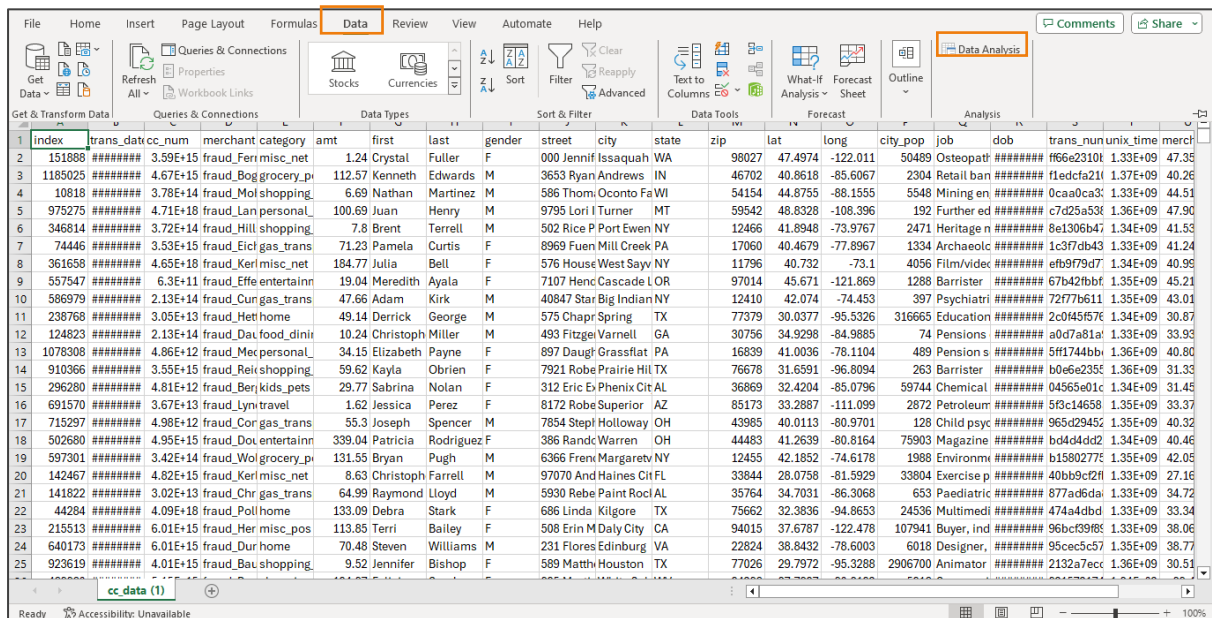


Capstone 1: Excel

Data Exploration

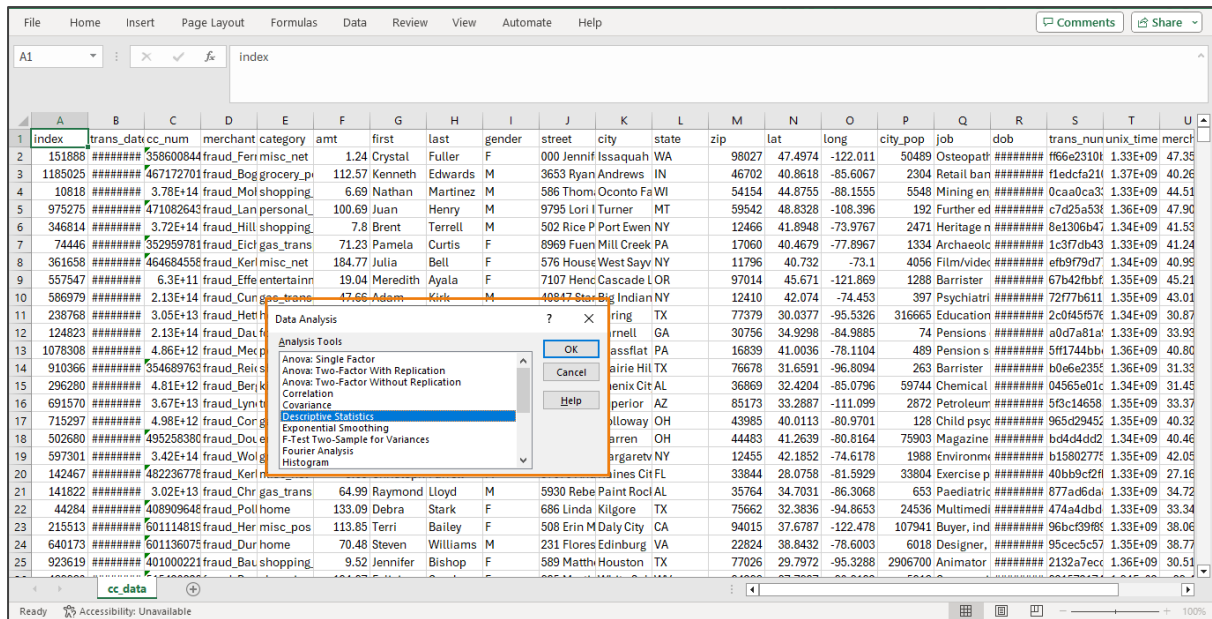
1. Create statistical summary on amt and city_pop

1.1 Go to the **Data** tab and click on **Data Analysis** (If it is not visible, you might need to enable it in Excel options.)

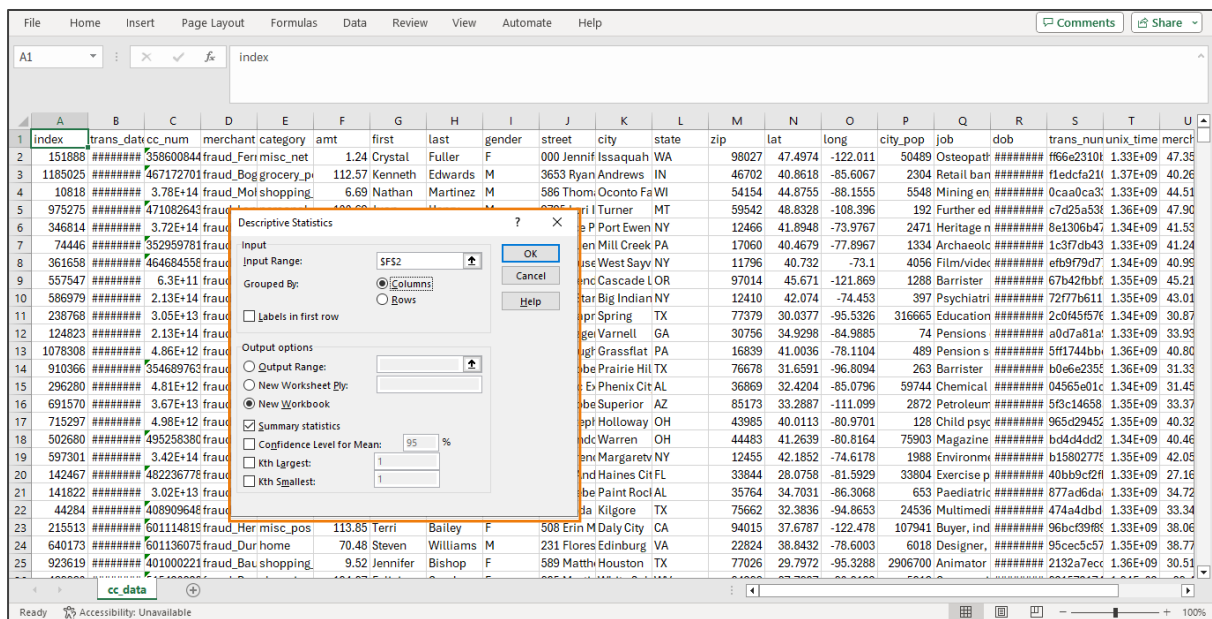


Index	trans_dat	cc_num	merchant category	amt	first	last	gender	street	city	state	zip	lat	long	city_pop	job	dob	trans_nun	unix_time	merch
1	151888	#####	3.59E+15 fraud_Ferri misc_net	1.24	Crystal	Fuller	F	000 Jennif Issaquah	WA		98027	47.4974	-122.011	50489	Osteopat	#####	f66e2310t	1.33E+09	47.35
2	1185025	#####	4.67E+15 fraud_Bog grocery_p	112.57	Kenneth	Edwards	M	3653 Ryan Andrews	IN		46702	40.8618	-85.6067	2304	Retail ban	#####	f1edcfa21i	1.37E+09	40.26
3	10818	#####	3.78E+14 fraud_Mol shopping	6.69	Nathan	Martinez	M	586 Thomi Oconto Fa WI			54154	44.8755	-88.1555	5548	Mining en	#####	0caa0ca3i	1.33E+09	44.51
4	975275	#####	4.71E+18 fraud_Lan personal	100.69	Juan	Henry	M	9795 Lori I Turner	MT		59542	48.8328	-108.396	192	Further ed	#####	c7d25a53i	1.36E+09	47.90
5	346814	#####	3.72E+14 fraud_Hill shopping	7.8	Brent	Terrell	M	502 Rice P Port Ewen NY			12466	41.8948	-73.9767	2471	Heritage n	#####	8e1306b4i	1.34E+09	41.53
6	74446	#####	3.53E+15 fraud_Eicl gas_trans	71.23	Pamela	Curtis	F	8969 Fuen Mill Creek PA			17060	40.4679	-77.8967	1334	Archaeotc	#####	1c3f7db4i	1.33E+09	41.24
7	361658	#####	4.65E+18 fraud_Kerl misc_net	184.77	Julia	Bell	F	576 House West Sayv NY			11796	40.732	-73.1	4056	Film/videt	#####	efb979d7i	1.34E+09	40.99
8	557547	#####	6.3E+11 fraud_Effe entertainn	19.04	Meredith	Ayala	F	7107 Henc Cascade L OR			97014	45.671	-121.869	1288	Barrister	#####	67b42fbfi	1.35E+09	45.21
9	586979	#####	2.13E+14 fraud_Cung gas_trans	47.66	Adam	Kirk	M	40847 Star Big Indian NY			12410	42.074	-74.453	397	Psychiatri	#####	72f77b61i	1.35E+09	43.01
10	238768	#####	3.05E+13 fraud_Het home	49.14	Derrick	George	M	575 Chapr Spring TX			77379	30.0377	-95.5326	316665	Education	#####	2c044567i	1.34E+09	30.87
11	124823	#####	2.13E+14 fraud_Dai food_dinir	10.24	Christoph	Miller	M	493 Fitzgel Varnell GA			30756	34.9298	-84.9885	74	Pensions	#####	a0d7a81ai	1.33E+09	33.93
12	1078308	#####	4.86E+12 fraud_Mec personal	34.15	Elizabeth	Payne	F	897 Daugt Grassflat PA			16839	41.0036	-78.1104	489	Pension s	#####	5f11744bbi	1.36E+09	40.80
13	910366	#####	3.55E+15 fraud_Rei shopping	59.62	Kayla	Obrien	F	7921 Robe Prairie Hil TX			76678	31.6591	-96.8094	263	Barrister	#####	b0e6e235i	1.36E+09	31.33
14	296280	#####	4.81E+12 fraud_Berj kids_pets	29.77	Sabrina	Nolan	F	312 Eric Ex Phenix Cit AL			36869	32.4204	-85.0796	59744	Chemical	#####	04565e01c	1.34E+09	31.45
15	691570	#####	3.67E+13 fraud_Lyn travel	1.62	Jessica	Perez	F	8172 Robe Superior AZ			85173	33.2887	-111.099	2872	Petroleum	#####	5f3c14658	1.35E+09	33.37
16	715297	#####	4.98E+12 fraud_Cor gas_trans	55.3	Joseph	Spencer	M	7854 Stepl Holloway OH			43985	40.0113	-80.9701	128	Child psyc	#####	965d2945i	1.35E+09	40.32
17	502680	#####	4.95E+15 fraud_Doi entertainn	339.04	Patricia	Rodriguez	F	386 Randc Warren OH			44483	41.2639	-80.8164	75903	Magazine	#####	b4d4dd2i	1.34E+09	40.46
18	597301	#####	3.42E+14 fraud_Vol grocery_p	131.55	Bryan	Pugh	M	6366 Fren Margaretv NY			12455	42.1852	-74.6178	1988	Environm	#####	b1580277i	1.35E+09	42.05
19	142467	#####	4.82E+15 fraud_Kerl misc_net	8.63	Christoph	Farrell	M	97070 And Haines Cit FL			33844	28.0758	-81.5929	33804	Exercise p	#####	40b49c2fi	1.33E+09	27.16
20	141822	#####	3.02E+13 fraud_Chr gas_trans	64.99	Raymond	Lloyd	M	5930 Rebe Paint Roc AL			35764	34.7031	-86.3068	653	Paediatric	#####	877ad5dai	1.33E+09	34.72
21	44284	#####	4.09E+18 fraud_Poll home	133.09	Debra	Stark	F	686 Linda Kilgore TX			75662	32.3836	-94.8653	24536	Multimed	#####	474a4dbd	1.33E+09	33.34
22	215513	#####	6.01E+15 fraud_Her misc_pos	113.85	Terri	Bailey	F	508 Erin M Daly City CA			94015	37.6787	-122.478	107941	Buyer, ind	#####	96bcf39f8i	1.33E+09	38.06
23	640173	#####	6.01E+15 fraud_Our home	70.48	Steven	Williams	M	231 Flores Edinburg VA			22824	38.8432	-78.6003	6018	Designer,	#####	95cec5c5i	1.35E+09	38.77
24	923619	#####	4.01E+15 fraud_Bau shopping	9.52	Jennifer	Bishop	F	589 Matthi Houston TX			77026	29.7972	-95.3288	2906700	Animator	#####	2132a7ecc	1.36E+09	30.51

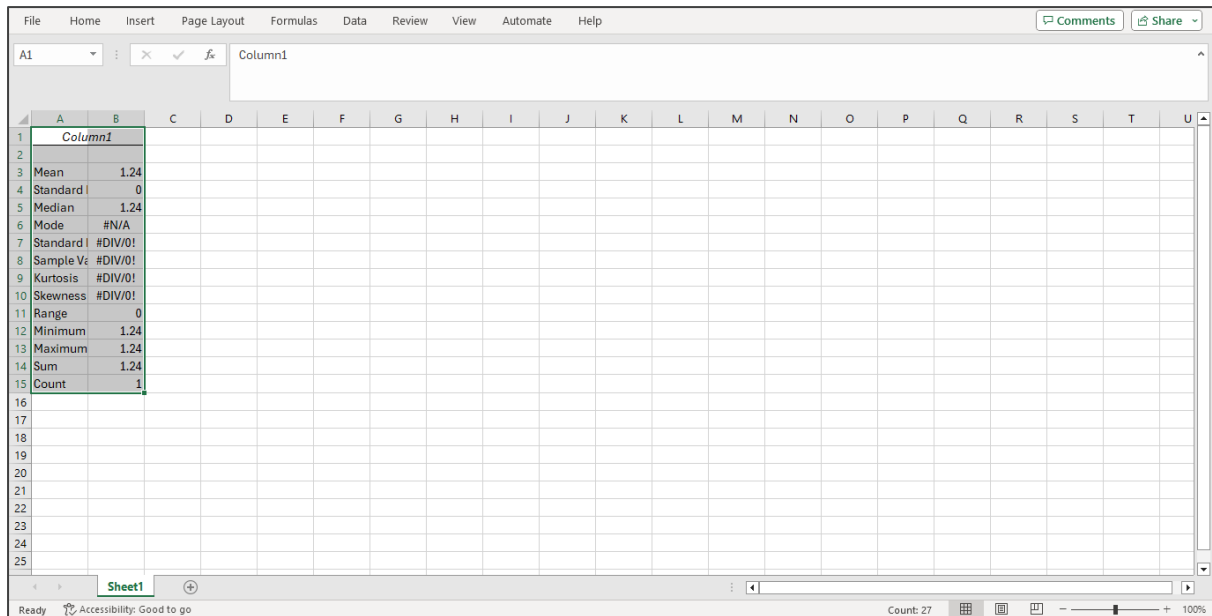
1.2 Select **Descriptive Statistics** from the list and click **OK**



1.3 Input the range for **amt**, check the **Summary statistics** option, select an output option as **New Workbook**, and click **OK**

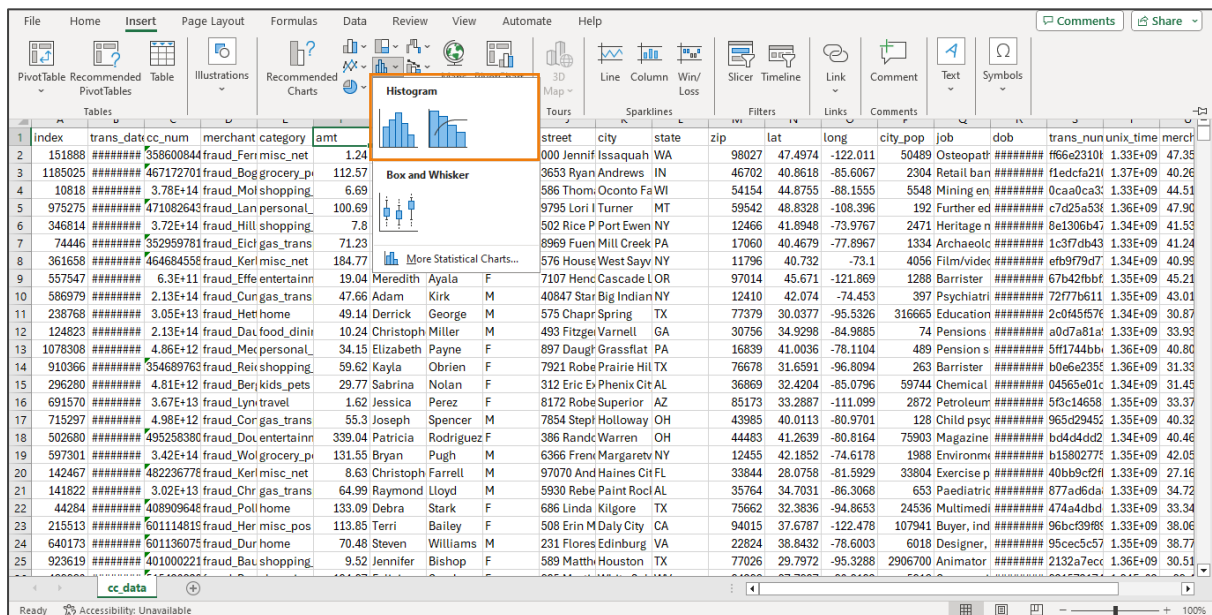


Here is the sample output for input range of column **amt** with data cell \$F\$2, which creates a new notebook. Same can be done for **city_pop** column.

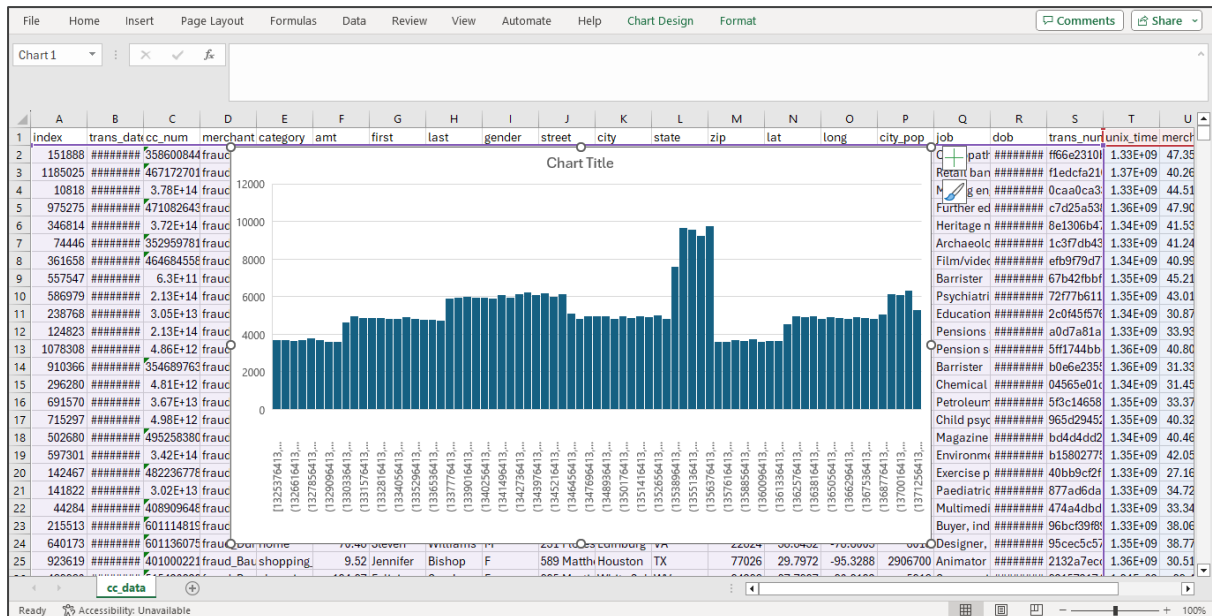


2. Plot histogram on amt

2.1 Select the **amt** column, go to the **Insert** tab, click on either **Insert Statistic Chart**, or choose **Histogram** from the chart options

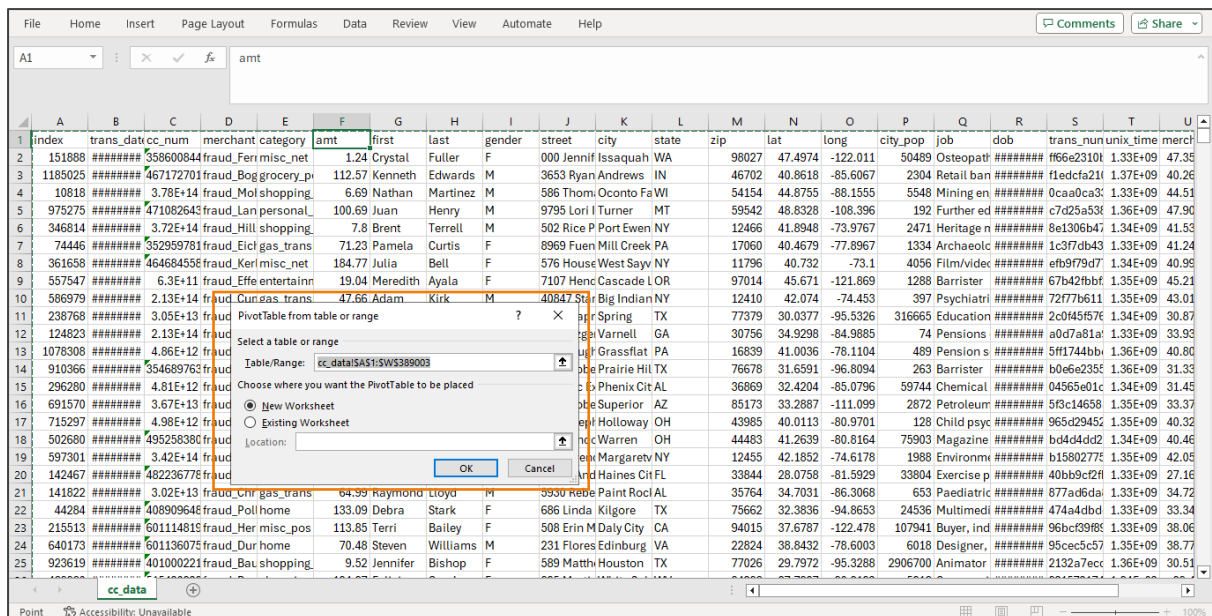


2.2 Adjust the bin size and other settings accordingly



3. Create a report showing number of Frauds by gender and category

3.1 Use a pivot table by selecting the entire dataset, going to the **Insert** tab, clicking on **PivotTable**, and choosing its placement



3.2 Drag **Gender** to the rows area, **Category** to the columns area, and the **Count of index** to the values area, then format the pivot table as required

The screenshot shows an Excel spreadsheet with a PivotTable. The PivotTable is located in the range A3:K7. The data is summarized by 'category' (Columns) and 'Sum of index' (Values). The categories listed are: entertainment, food_dining, gas_transport, grocery_net, grocery_pos, health_fitness, home, kids_pets, misc_net, and misc_pos. The 'Sum of index' values are: 18315406100, 17429716564, 25757920811, 8950852022, 23912161667, 16684129092, 23923857777, 22090189064, 12446823304, and 155487976.

The PivotTable Fields task pane is open on the right, showing the configuration:

- Filters:** (Empty)
- Columns:** category
- Rows:** gender
- Values:** Sum of index

4. Show top 3 states by highest number of transactions

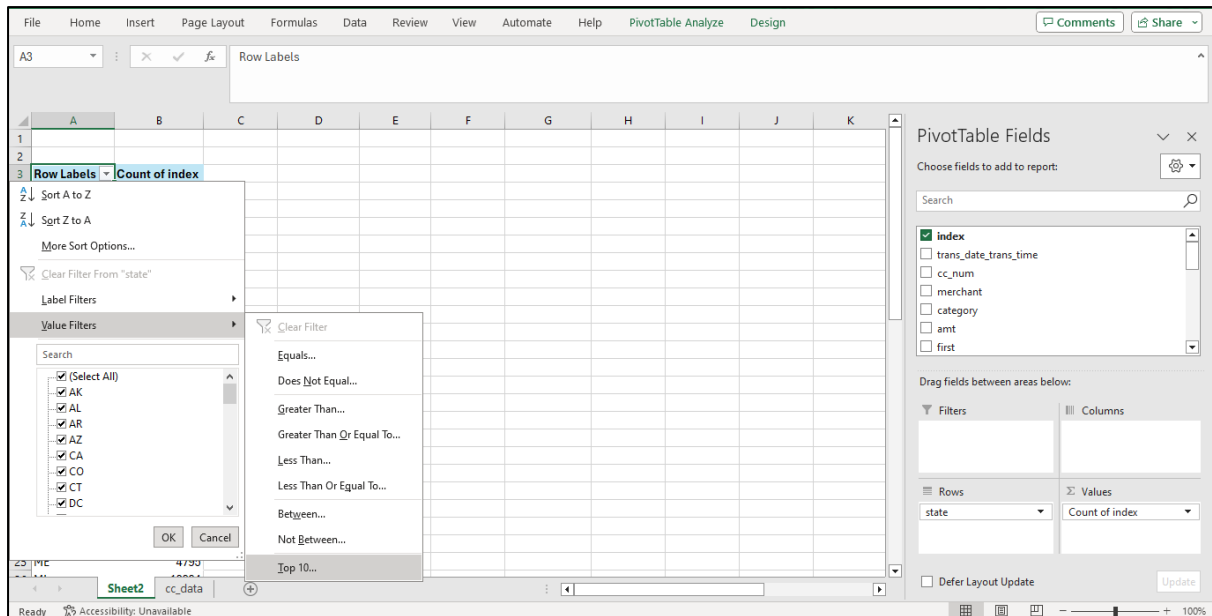
4.1 Use a pivot table by selecting the entire dataset, accessing the **Insert** tab, clicking on **PivotTable** to choose its placement, and then dragging **State** to the rows area and the **Count of index** to the values area

The screenshot shows an Excel spreadsheet with a PivotTable. The PivotTable is located in the range A3:K7. The data is summarized by 'state' (Rows) and 'Count of index' (Values). The states listed are: AK, AL, AR, AZ, CA, CO, CT, DC, DE, FL, GA, HI, IA, ID, IL, IN, KS, KY, LA, MA, MD, and ME. The 'Count of index' values are: 597, 12321, 9338, 3243, 17007, 4110, 2380, 1061, 1, 12896, 7912, 757, 8101, 1710, 12863, 8200, 6858, 8601, 6293, 3718, 7853, and 4795.

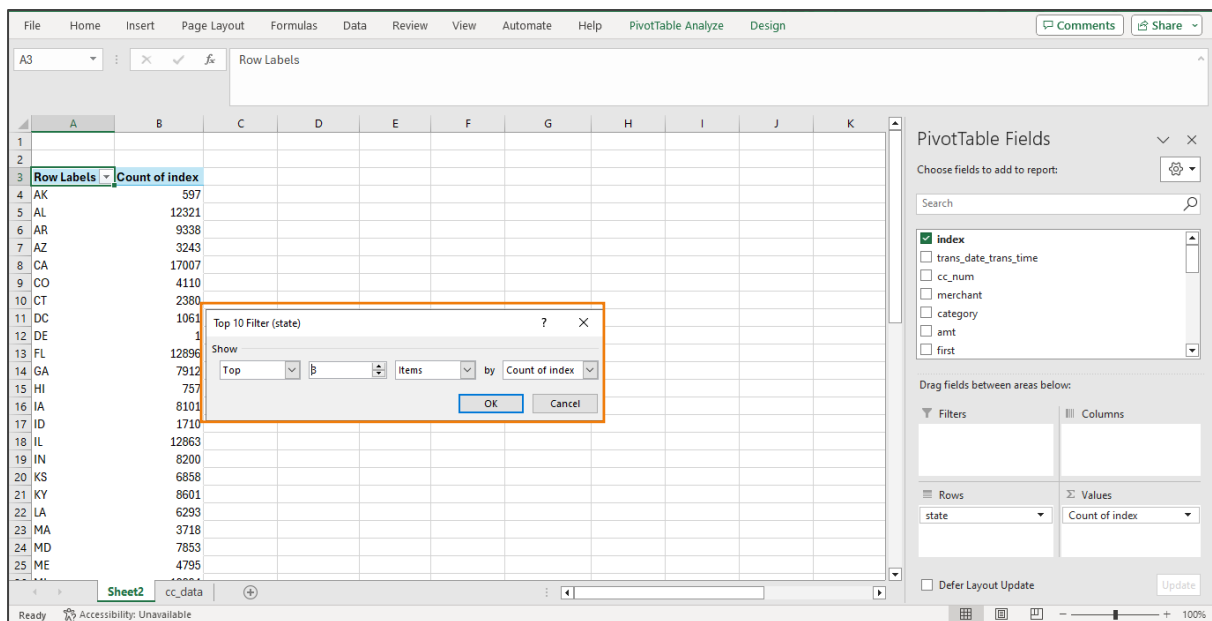
The PivotTable Fields task pane is open on the right, showing the configuration:

- Filters:** (Empty)
- Columns:** (Empty)
- Rows:** state
- Values:** Count of index

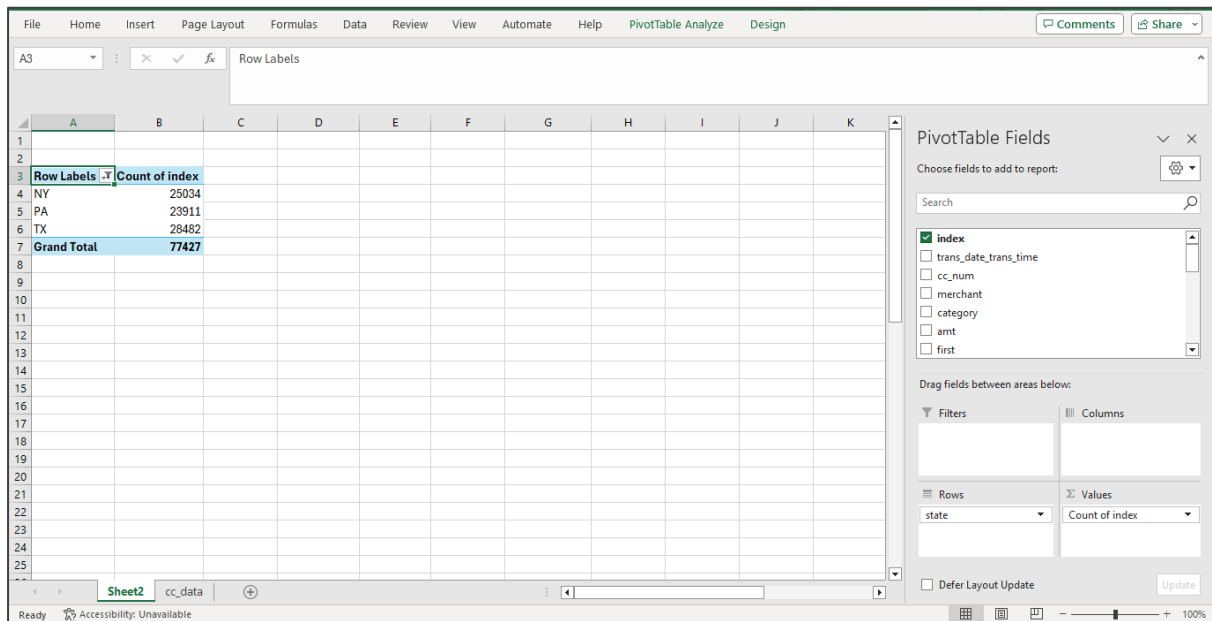
4.2 Filter the pivot table by clicking on Row Labels, selecting **Value Filters**, and then choosing **Top 10**



4.3 Select **Top, 3, Items**, and by as **Count of index** in the **Top 10 Filter (state)** window



Here are the top 3 states by highest number of transactions.

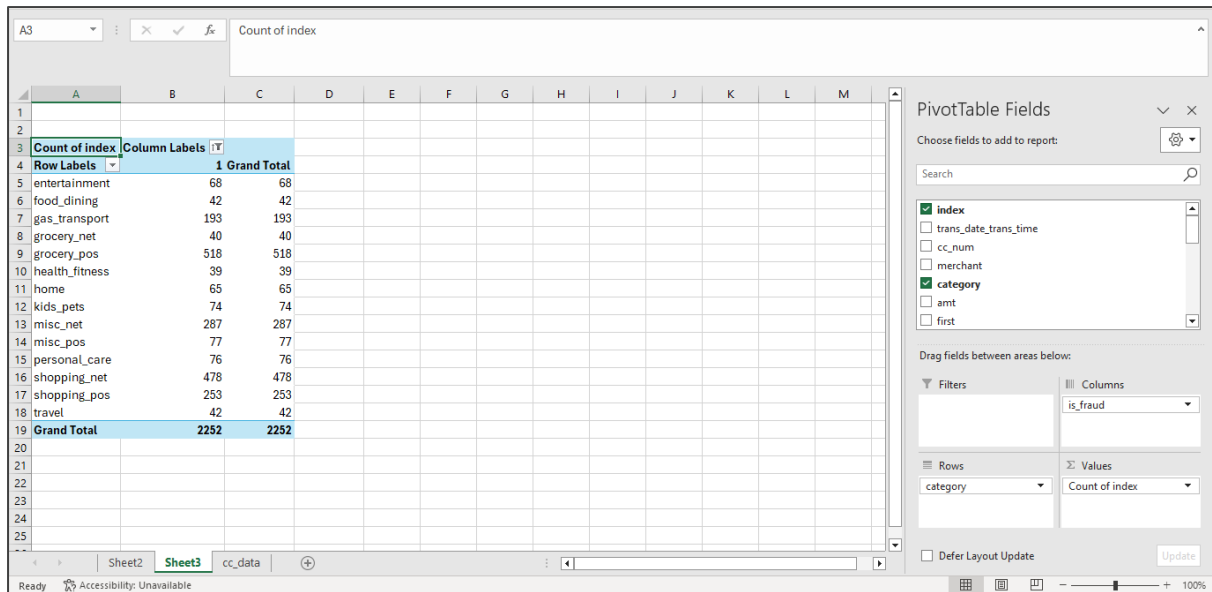


Data Analysis in Excel

1. Is there a correlation between transaction amount and city population?
Yes
2. How many fraudulent transactions occurred in each category?
 - 2.1 Create a PivotTable by selecting the entire dataset, including the **Category** and **is_fraud** columns, then go to the **Insert** tab, click on **PivotTable**, choose its placement, and click **OK** to finalize
 - 2.2 Drag the **Category** field to the rows area, the **is_fraud** field to the column area, and **index** into the value field, selecting count as the aggregate function

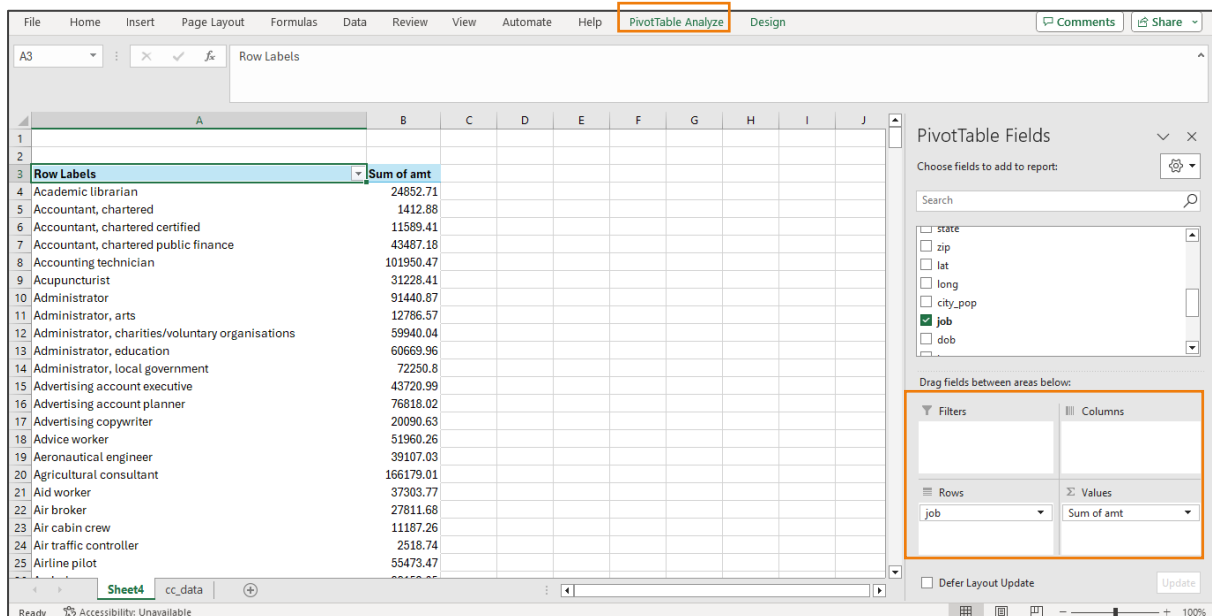
2.3 To filter only fraudulent transactions, click on the arrow next to **Column Labels** in the PivotTable, and then check only 1

Here, it results in displaying only the count of fraudulent transactions for each category in the pivot table.

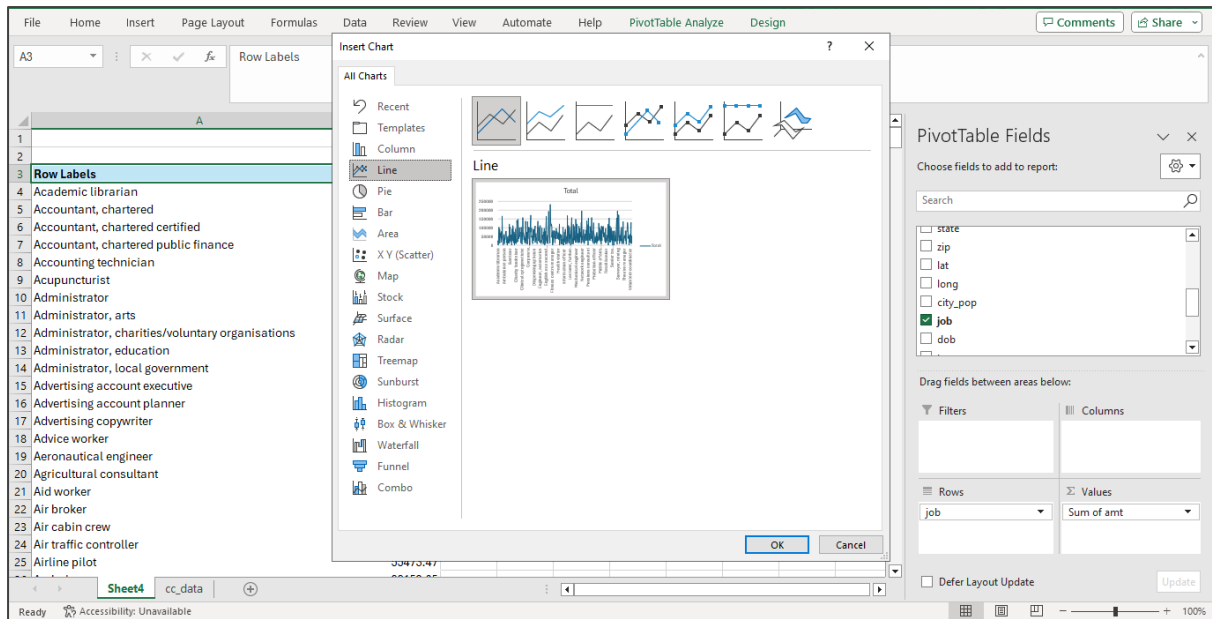


3. How does the average transaction amount vary between different job roles?

3.1 Create a pivot table, dragging **job** role to the rows area and **amt** to the values area, then format the pivot table as necessary



3.2 In the pivot table screen, click on **PivotTable Analyze**, choose **Pivot Charts** to create a line chart



Hence the chart is created.

